

# JEE MAIN REVISION MASTER: RAY OPTICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 38 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                     | Frequency | Category             |
|---------------|--------------------------------------------------|-----------|----------------------|
| Very High     | Reflection at Spherical Mirrors & Mirror Formula | 17 PYQs   | Mirrors              |
| High          | Refraction at Interfaces, Snell's Law & TIR      | 12 PYQs   | Refraction           |
| High          | Lens Maker's Formula & Optical Instruments       | 11 PYQs   | Lenses & Instruments |
| Moderate      | Prism Dispersion & Minimum Deviation             | 7 PYQs    | Prisms               |

## 2. Master Formula Reference Card

| Concept / Topic                                  | Mathematical Formulation & Governing Equations                                                                                                                                                |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reflection at Spherical Mirrors & Mirror Formula | $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ , $m = -\frac{v}{u} = \frac{f}{f - u}$ , $f = \frac{R}{2}$                                                                                          |
| Refraction at Interfaces, Snell's Law & TIR      | $n_1 \sin i = n_2 \sin r$ , $\sin \theta_c = \frac{n_2}{n_1}$ ; ( $n_1 > n_2$ ), $d' = \frac{d}{\mu}$ , $\text{Shift} = t \left( 1 - \frac{1}{\mu} \right)$                                   |
| Lens Maker's Formula & Optical Instruments       | $\frac{1}{f} = (\mu - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$ , $P = \frac{1}{f}$ , $M_{\text{telescope}} = \frac{f_o}{f_e}$ , $M_{\text{microscope}} = \frac{L}{f_o} \frac{D}{f_e}$ |
| Prism Dispersion & Minimum Deviation             | $\delta = i + e - A$ , $\mu = \frac{\sin \left( \frac{A + \delta_m}{2} \right)}{\sin(A/2)}$ , $\delta \approx (\mu - 1)A$ ; (thin prism)                                                      |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Reflection at Spherical Mirrors & Mirror Formula (Very High Yield — 17 PYQs)

**Core Principle:** Image formation by concave and convex mirrors, Cartesian sign convention, and linear/longitudinal magnification.

■ **Caution / JEE Pitfall:** Concave mirror has negative focal length ( $f < 0$ ); convex mirror has positive focal length ( $f > 0$ ).

■ **Speed Trick:** Virtual image in a concave mirror is ALWAYS magnified ( $|m| > 1$ ), while in a convex mirror it is ALWAYS diminished ( $|m| < 1$ ).

**Representative Source PYQs:**

- [2026 Shift] Q#4 (MCQ): "The direction of a ray of light incident on a concave mirror is shown by PQ, while direction in which the ray would travel after reflection ..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "An equiconvex lens of radius of curvature 14 cm is made up of two different materials. Left half and right half of vertical portion is made ..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The final image formed by an astronomical telescope is..." [Ans: N/A]

### 3.2 Refraction at Interfaces, Snell's Law & TIR (High Yield — 12 PYQs)

**Core Principle:** Snell's law, apparent depth shift in transparent media, and total internal reflection condition ( $i \geq \theta_c$  from denser to rarer).

■ **Caution / JEE Pitfall:** Total internal reflection can ONLY occur when light travels from an optically DENSER medium to a RARER medium.

■ **Speed Trick:** Area of circle through which light emerges from fish/source at depth  $h$ :  $A = (\pi h^2 / \mu^2 - 1)$ .

**Representative Source PYQs:**

- [2026 Shift] Q#2 (MCQ): "The critical angle for a monochromatic light going from medium A to medium B is  $\theta$ . If the speed of light in medium A is  $V$ , then t..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "The incorrect statement about refractive index for a pair of media is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "A ray of light passes from vacuum into a medium of refractive index  $n$ . If the angle of incidence is twice the angle of refraction, then t..." [Ans: N/A]

### 3.3 Lens Maker's Formula & Optical Instruments (High Yield — 11 PYQs)

**Core Principle:** Thin lens refraction, Lens Maker's equation, lens power in Dioptres ( $P = 1/f$ ), and magnifying power of optical devices.

■ **Caution / JEE Pitfall:** When a lens is immersed in a liquid of refractive index  $\mu_l > \mu_{\text{(glass)}}$ , its converging/diverging nature REVERSES (convex acts concave).

■ **Speed Trick:** When a lens is cut vertically in half, each half retains the SAME focal length  $f$ . When cut horizontally, focal length remains  $f$ , but image brightness is halved.

**Representative Source PYQs:**

- [2025 Shift] Q#3 (MCQ): "A convex lens has power  $P$ . It is cut into two halves along its principal axis. Further one piece (out of two halves) is cut into two halves ..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "The image formed by an objective lens of a compound microscope is..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "An equiconvex lens made of glass of refractive index  $(3/2)$  has focal length  $f$  in air. It is completely immersed in water of refrac..." [Ans: N/A]

### 3.4 Prism Dispersion & Minimum Deviation (Moderate Yield — 7 PYQs)

**Core Principle:** Ray refraction through triangular prisms, condition for minimum deviation ( $i = e$ ,  $r_1 = r_2 = A/2$ ), and angular dispersion.

■ **Caution / JEE Pitfall:** At minimum deviation, the refracted ray inside an equilateral prism passes strictly PARALLEL to the base.

■ **Speed Trick:** Condition for no emergent ray (always TIR on second face):  $A > 2\theta_c$ .

**Representative Source PYQs:**

- **[2026 Shift] Q#1 (MCQ):** "From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is..." **[Ans: N/A]**
- **[2026 Shift] Q#1 (MCQ):** "With reference to the figure shown below, match the following: List-I List-II (a) Angle of reflection (i)  $60^\circ$  (b) Value of  $\alpha$ ..." **[Ans: A]**
- **[2025 Shift] Q#1 (MCQ):** "If  $r$  and  $r^1$  denotes the angles inside the prism having angle of prism  $50^\circ$  considering that during interval of time from  $t=0$ ..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

From the graph of angle of deviation versus angle of incidence for an equilateral prism, the refractive index of material of prism is

- (A)  $\frac{\sqrt{3}}{2}$
- (B)  $\frac{3}{2}$
- (C)  $\sqrt{3}$
- (D)  $\sqrt{2}$

**Correct Answer: None**

**Solution Analysis:** For an equilateral prism, the angle of prism is  $A=60^\circ$ . From the graph of angle of deviation  $\delta$  versus angle of incidence  $i$ , the lowest value of deviation is the minimum deviation. Here, the graph shows  $\delta_m=60^\circ$ . The refractive index  $\mu$  of the prism material is given by  $\mu = \frac{\sin(A + \delta_m/2)}{\sin(A/2)}$ . Now put  $A=60^\circ$  and  $\delta_m=60^\circ$  in the formula.  $\mu = \frac{\sin(60^\circ + 60^\circ/2)}{\sin(60^\circ/2)}$   $\mu = \frac{\sin(90^\circ)}{\sin(30^\circ)}$  Using  $\sin 60^\circ = \frac{\sqrt{3}}{2}$  and

### Benchmark Problem 2: KCET 2026 (Q#2)

The critical angle for a monochromatic light going from medium A to medium B is  $\theta$ . If the speed of light in medium A is  $V$ , then the speed of light in medium B is

- (A)  $V(1 - \cos\theta)$
- (B)  $(V / \cos\theta)$
- (C)  $(V / \sin\theta)$  **[Correct]**
- (D)  $V(1 - \sin\theta)$

**Correct Answer: C**

**Solution Analysis:** For critical angle, light must be travelling from a denser medium to a rarer medium. Here, light goes from medium A to medium B, and the critical angle is  $\theta$ . At the critical angle, the angle of refraction in medium B becomes  $90^\circ$ . Using Snell's law,  $n_A \sin\theta = n_B \sin 90^\circ$ . Since  $\sin 90^\circ = 1$ ,  $n_A \sin\theta = n_B$ . Now, refractive index is related to speed of light by  $n = (c / v)$ . So,  $n_A = (c / V)$ . Let the speed of light in medium B be  $v_B$ . Then,  $n_B = (c / v_B)$ . Substituting in the equation,  $(c / V) \sin\theta = (c / v_B)$ . Cancel

**Benchmark Problem 3: KCET 2026 (Q#4)**

The direction of a ray of light incident on a concave mirror is shown by PQ, while direction in which the ray would travel after reflection is shown by four rays marked as A, B, C and D as shown in the figure. Which of the four rays correctly shows the direction of the reflected ray?

- (A)
- (B)
- (C)
- (D)

**Correct Answer: None**

**Solution Analysis:** For a concave mirror, there is an important rule: a ray of light which is parallel to the principal axis reflects and then passes through the principal focus,  $F$ . In the figure, the incident ray PQ is parallel to the principal axis of the concave mirror. So after reflection, this ray must pass through the focus  $F$ . Among the given rays, only ray C passes through the principal focus  $F$ . So, the correct reflected ray is C.

**Benchmark Problem 4: KCET 2025 (Q#4)**

The image formed by an objective lens of a compound microscope is

- (A) Real and diminished
- (B) Real and enlarged
- (C) Virtual and enlarged
- (D) Virtual and diminished

**Correct Answer: None**

**Solution Analysis:** The objective lens of a compound microscope produces a real, inverted and magnified image of the object. So the correct choice is: Option B: Real and enlarged